

BloomBelly

AI-Assisted, Arabic-First Maternal & Child Health Platform

Graduation Thesis · College of Informatics Engineering · Al-Sham Private University (ASPU) · Damascus, Syria · 2026

The Problem. Maternal and child mortality in Arabic-speaking regions is among the world's highest. Most digital health apps are English-first; translations are mechanical; fathers have no clear role; information is contradictory. This documentably increases anxiety and delays detection of warning signs during the highest-leverage health window known: the "first 1,000 days."

The Solution. BloomBelly unifies pregnancy tracking, AI medical image analysis, evidence-based fetal movement monitoring, nutrition evaluation, and a dedicated father module under one Arabic-first interface. Three specialized AI components — Gemini for medical images, a LoRA-tuned transformer for the chatbot, and a Random Forest classifier for nutrition — operate behind a Python Flask backend with Supabase persistence.

Evidence-Based Foundation

Feature	Reference
Kick counter (ACOG-aligned)	Saastad et al., PLOS ONE (2011, RCT, n=1076)
Reduced movement awareness	Weller et al., PLOS Global Public Health (2023)
Partner support	Versele et al., Frontiers in Public Health (2022)
Maternal nutrition	WHO Guidelines · King (2016)
Pediatric sleep	Paruthi et al., AASM (2016)
Vaccination schedules	Syrian Ministry of Health · WHO EPI

■ SDG Alignment • SDG 3 — Good Health (3.1, 3.2, 3.7, 3.8) • SDG 5 — Gender Equality (5.6, 5.b) • SDG 10 — Reduced Inequalities (10.2) • SDG 4 — Quality Education (4.7)	■ Market • TAM: ~5.6M annual pregnancies (Arab League, smartphone-equipped) • SAM: \$8M–\$15M (Levant + Gulf + Egypt) • SOM (3 yr): \$1.6M–\$4.5M • Distribution: Clinic partnerships	■ Team • Mozn Jamous — Backend, AI, architecture • Shahd Bureghsh — Frontend, UX, product • Dr. Afaf Al-Shalabi — Principal Supervisor • Eng. Rahaf Abdul Qader — Technical Supervisor
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Technology Stack

Layer	Choice	Why
Mobile	Flutter (MVVM + Provider)	Single codebase, mature Arabic typography
Backend	Python Flask	Native ecosystem for AI orchestration
Medical AI	Google Gemini API	Strong multimodal image analysis
Chatbot	LoRA fine-tuned transformer	Lightweight domain specialization
Nutrition	Random Forest (scikit-learn)	Interpretable, deterministic, fast
Database	Supabase (PostgreSQL + RLS)	Realtime sync, robust auth, free tier
Auth	JWT + bcrypt	Stateless, industry-standard

We are actively seeking:

Digital health grants (Grand Challenges, UNICEF, WHO) · Accelerator partnerships (Plug and Play Health, Flat6Labs, Misk, MBRF) · Research collaborations on Arabic medical NLP · Letters of support and pilot clinics in the Levant and Gulf regions.

Maternal and child health is the single highest-leverage investment any society can make. BloomBelly is our contribution to that investment, built in the region we know best, for the population that has been underserved by global digital health to date.